

Distilling Multi-modal Large Language Models for Autonomous Driving

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Abstract

Autonomous driving demands safe motion planning, especially in critical “long-tail” scenarios. Recent end-to-end autonomous driving systems leverage large language models (LLMs) as planners to improve generalizability to rare events. However, using LLMs at test time introduces high computational costs. To address this, we propose DiMA, an end-to-end autonomous driving system that maintains the efficiency of an LLM-free (or vision-based) planner while leveraging the world knowledge of an LLM. DiMA distills the information from a multi-modal LLM to a vision-based end-to-end planner through a set of specially designed surrogate tasks. Under a joint training strategy, a scene encoder common to both networks produces structured representations that are semantically grounded as well as aligned to the final planning objective. Notably, the LLM is optional at inference, enabling robust planning without compromising on efficiency. Training with DiMA results in a 37% reduction in the L2 trajectory error and an 80% reduction in the collision rate of the vision-based planner, as well as a 44% trajectory error reduction in long-tail scenarios. DiMA also achieves state-of-the-art performance on the nuScenes planning benchmark.

1. Introduction

Research in autonomous driving has moved away from the traditional strategy of integrating sequential, independently trained models [4, 22, 26, 29, 31, 45, 52], and moved towards developing multi-task systems that are trained in an end-to-end manner [3, 5, 7, 16, 20, 49]. These approaches demonstrate improved performance and interpretability while being highly efficient. However, they struggle with long-tail navigation and perception scenarios, primarily because they are trained on task-specific, limited datasets [2, 11, 13].

Large language models (LLMs) have emerged as a promising solution to this issue. Trained on vast, internet-

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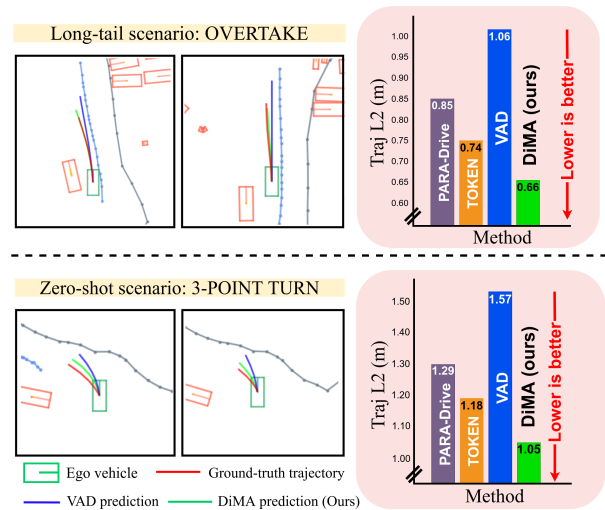


Figure 1. Comparison of planning performance in long-tail scenarios from nuScenes: DiMA-VAD demonstrates greater robustness compared to VAD [20] in long-tail navigation scenarios such as overtaking a vehicle and performing a 3-point turn. DiMA-VAD also outperforms recent vision-planner PARA-Drive [48] and LLM planner TOKEN [38]. Notably, the 3-point turn is a zero-shot scenario that is only present in the validation set.

scale datasets, LLMs can leverage world knowledge to generalize to unseen or rare scenarios. These models can perform high-level reasoning tasks using mechanisms such as chain-of-thought [46]. Going beyond text-based prompting, multi-modal large language models (MLLMs) integrate image and video inputs to the LLM, enabling tasks such as visual question answering and dense captioning [21, 24]. Recent end-to-end autonomous driving systems leverage LLMs to achieve higher interpretability through language-guided reasoning and superior robustness to long-tail scenarios [19, 37–39, 43, 44]. To differentiate between these methods, we call end-to-end planners that depend on LLMs to perform trajectory prediction as “LLM-based” planners and ones that do not as “vision-based” planners [18, 20, 48]. In spite of their recent success “LLM-based” planners face significant challenges.

Firstly, LLM-based planners require a significant amount of computational overhead at test time, limiting their practicality. This work addresses a core question: *how can we harness LLMs’ world knowledge while preserving the efficiency of vision-based planners?* Secondly, bridging the visual and language domains is inherently more complex for end-to-end planning tasks compared to general MLLM tasks [21, 24], with the additional hurdle of limited training data. Standard image tokenization strategies rely on frozen pre-trained image encoders [33] to generate dense, unstructured multi-modal token embeddings. [21, 24, 37]. We suggest that MLLMs meant for end-to-end autonomous driving benefit from structured inputs that explicitly model scene components. Furthermore, we suggest that updating these features during training results in richer representations.

To address these challenges, we propose **DiMA**, a novel framework for **Distilling Multi-modal Large Language Models for Autonomous driving**. We introduce a joint-training scheme between a vision-based planner and an MLLM that enables the learning of robust, grounded and disentangled scene representations that are aligned to the final objective of planning. Specifically, we use the vision-based planner as a tokenizer, and pass the learned scene representations to the MLLM, providing a more structured input. The MLLM is then trained for the tasks of visual question-answering, trajectory estimation, and a series of surrogate tasks designed to ground the multi-modal inputs in language. The vision-based planner is simultaneously trained for trajectory estimation, while also distilling features from the MLLM planning head to the planning transformer. Crucially, **the MLLM can be discarded when performing planning inference**, maintaining the efficiency of the vision-based planner while utilizing the knowledge of the language model. Optionally, the MLLM can support language-guided reasoning through visual question answering. Our experiments show that DiMA is more robust to challenging long-tail scenarios compared to baselines and state-of-the-art methods [38, 48]. Figure 1 confirms this both quantitatively and qualitatively. We list the contributions of the work below:

1. We introduce **DiMA**, an end-to-end autonomous driving framework that distills knowledge from an MLLM to a vision-based planner to ensure robustness to long-tail events while maintaining efficiency. DiMA is capable of planning as well as visual question-answering.
2. We propose a distillation task along with the following surrogate tasks to align the objectives of the vision-based planner and the MLLM: (i) masked token reconstruction (ii) future token prediction (iii) scene editing.
3. DiMA outperforms both vision-based and LLM end-to-end planners, achieving state-of-the-art results on the nuScenes planning benchmark. Training with DiMA results in a 37% reduction in the L2 trajectory error and an

80% reduction in the collision rate of the vision-based planner, as well as a 44% trajectory error reduction in long-tail scenarios.

2. Related Work

2.1. End-to-End Autonomous Driving

Autonomous driving (AD) systems are generally modular, consisting of tasks such as perception [22, 31, 45, 52], motion prediction [4, 26, 29], and planning [9]. Classic AD systems use standalone models for each task where each module is optimized separately and combined sequentially. Alternatively, some approaches incorporate these modular tasks into a multi-task learning paradigm, where tasks shared a common feature extraction process but have task-specific heads [8, 23, 53, 55]. Recently, research has moved toward optimizing autonomous driving systems in an end-to-end manner, aiming to directly predict future trajectories [3, 5, 7, 16, 20, 49]. Along this line of work, UniAD [18] introduces a query-based design that integrates perception and prediction components, enabling an end-to-end planning scheme. VAD [20] utilizes a vectorized scene representation, replacing the dense rasterized representations from [18], reducing computational cost and improving planning performance. While effective in general navigation scenarios, these methods struggle in difficult long-tail events (see Figure 1). In this work, we propose a method for end-to-end autonomous driving that is robust to such scenarios.

2.2. LLMs for Autonomous Driving

The recent success of large language models [12, 41, 42] demonstrates their ability to process complex contextual information with logical reasoning and to communicate in a human-interpretable manner, attracting significant attention in autonomous driving [6, 10, 14, 25, 28, 35–37, 39, 44, 47, 50]. These works investigate the capabilities of language models to generalize to novel scenarios as well as their ability to reason about the scene in the form of text.

Prompting LLMs. One category of approaches involves prompting LLMs with text using reasoning frameworks such as chain-of-thought [46]. LanguageMPC [35] leverages the common-sense reasoning capacity of LLM to make high-level driving decisions, which are then translated into bottom-level control signals. GPT-Driver [28] converts observations and ego-states into language prompts, framing driving planning as a text-based task for LLMs.

End-to-end Driving with MLLMs. Another approach is to integrate multi-modal LLMs into end-to-end train frameworks for autonomous vehicles. LMDrive [36] integrates a vision encoder with a large language model, enabling natural language instruction in autonomous driving. DriveLM introduces multi-step question answering, going beyond single-round interactions [27, 32, 34]. DriveVLM [39] leverages MLLMs and a chain-of-thought process to

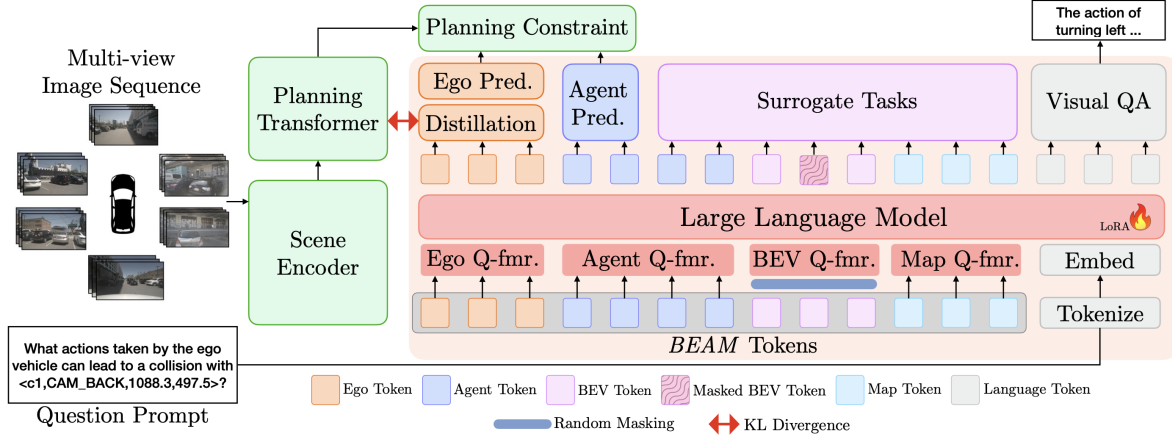


Figure 2. Overview of DiMA. The input to the framework is a multi-view image sequence and a question text prompt. The **vision-based end-to-end planner** consists of a scene encoder and a planning transformer. The scene encoder learns structured latent representations in the form of **bird’s-eye-view**, **ego**, **agent**, and **map** (*BEAM*) token embeddings and acts as a trainable tokenizer for the **multi-modal large language model** (MLLM). The planning transformer is trained under standard planning constraints [18, 20]. The MLLM is trained for planning, visual question answering, distillation, and a series of surrogate tasks.

enhance scene understanding and planning.

We point out a challenge in the current MLLM strategy for autonomous driving. We suggest that in order to reason about vehicle dynamics, the language model benefits from a structured input that explicitly models scene components instead of dense, unstructured input to the MLLM as in [36, 39, 43, 44]. In TOKEN [38], the scene encoder from [48] is used as a tokenizer. However, it is frozen during training. In our approach, the input to the MLLM is a set of latent representations learned from a pre-trained scene encoder that is *jointly trained with the MLLM*, allowing it to more learn grounded feature representations

3. DiMA Framework

We present DiMA, a framework for end-to-end autonomous driving. Given a sequence of multi-view images, the overall objective is to predict the future trajectory of the ego vehicle and answer questions about the scene. An overview of DiMA is shown in Figure 2. The framework has two main components: 1) the vision-based planner; 2) a multi-modal large language model consisting of adapter layers, an LLM, and a series of task-specific decoder heads.

3.1. Vision-based Planner

Vision-based end-to-end planners are trained in a multi-task fashion to perform perception, mapping, motion prediction and planning [18, 20, 40, 48]. Considering standard architecture design from [18, 20, 48] we decompose the planner into the scene encoder and the planning transformer. The scene encoder provides structured latent representations to the planning transformer that performs waypoint prediction. We call these representations as scene token embeddings. In our framework, we leverage the vision-based planner in

two ways. First, it acts as the primary network through which planning is performed, allowing for fast prediction at test time. Second, the scene encoder is shared with the MLLM, acting as tokenizer to provide it with a highly structured set of inputs. An important distinction of our framework from previous works [19, 38] is that the scene encoder is trained jointly with the MLLM.

Scene encoder The scene encoder models the scene as high-dimensional token embeddings that explicitly represent components such as the environment map, the ego vehicle, and the surrounding agents. This is achieved by introducing task specific, learnable query features that are supervised by a series of planning constraints [20]. First, a visual backbone encodes the input multi-view image sequence. The obtained feature map is projected into the bird’s-eye-view (BEV) space using a set of queries to obtain BEV token embeddings (B) [22]. The network learns structured representations of the map and agents in the form of map token embeddings (M) and agent token (A) embeddings by cross-attending the BEV features with map and agent queries. A randomly initialized learnable embedding is trained to learn the interaction of the ego vehicle with the agent and map components which is the ego query (E). We denote the set of these latent scene representations as *BEAM* token embeddings. These token embeddings are sent to a planning head that predicts future trajectories of the ego and agent vehicles. They are also used as input to the MLLM as multi-modal inputs, as discussed next.

3.2. Multi-modal LLM

Our objective is to distill knowledge from an MLLM to the vision-based planner. Towards this, we jointly train an MLLM with the vision-based planner. Specifically, the

scene encoder is used as a trainable tokenizer for the MLLM to generate the *BEAM* token embeddings, as well as used as an encoder for the planning transformer. The advantage from this strategy is two-fold. First, the MLLM receives a highly structured input that captures rich spatio-temporal information relevant to the autonomous driving task. Second, the scene encoder learns features that are grounded in language, improving the robustness of vision-based planning. The main components of the MLLM are adapter layers, the large language model, and a series of task-specific decoder heads which we discuss next.

3.2.1. Adaptation of Scene Tokens

In order to project the *BEAM* token embeddings efficiently while maintaining their distinctiveness, we leverage the query-transformer (Q-former) module [17] to compress the visual tokens before being input to the LLM. We implement component-specific Q-former layers, namely a Map Q-former, a Bird's-Eye-View (BEV) Q-former, an Ego Q-former, and an Agent Q-former. Each of these adapters transforms the representing different modalities into a sequence of fixed-length, higher-dimensional tokens, preparing them for processing by the MLLM.

3.2.2. MLLM Supervision

We design tasks for the MLLM with the objectives of, a) enriching the intermediate scene representations; b) grounding the scene token embeddings in language; and c) training the LLM for planning-related reasoning. In detail, the MLLM is trained for visual question answering, trajectory estimation, feature distillation, and a set of surrogate tasks [1, 51].

Visual question answering. The MLLM is trained for visual question answering (VQA) on question-answer pairs of 4 types: perception of the scene, prediction of the agent behavior, prediction of ego-vehicle behavior, and future planning steps [37]. The answer is predicted based on a multi-modal prompt consisting of the question embeddings and the projected *BEAM* token embeddings. This is constructed as a standard next-token prediction task and the VQA branch is supervised with a cross-entropy loss as in [21]. We denote the loss from this task head as $\mathcal{L}_{LLM}$.

Surrogate tasks: The main goal of joint training is to enrich the *BEAM* scene representations learned by the scene encoder. We design MLLM surrogate tasks whose objectives are in line with that of future trajectory prediction.

1) *Masked token reconstruction:* Each type of token embedding contributes to a holistic representation of the scene. In order to enrich the visual representations, we ask the network to reconstruct a masked BEV input based on the context present in the rest of the multi-modal sequence. We perform random masking after scene encoding and pass the token embeddings to the MLLM. A reconstruction head takes the latent representations from the penultimate layer of LLM and predicts reconstructed BEV token embeddings

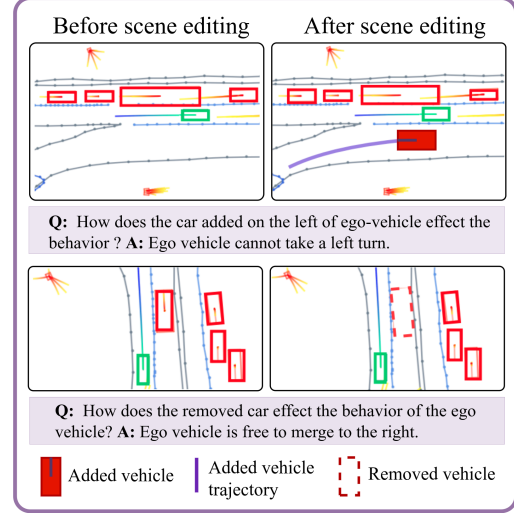


Figure 3. Examples of addition and deletion in scene editing. In the top row, a car (solid red box) is added on the left of the ego-vehicle (green box). In the bottom row, a car (dashed red box) is removed from the right of the ego-vehicle. A corresponding question-answer pair is created to characterize the edit.

$\hat{B}$. This decoder head is supervised with the L2 loss between the prediction and the complete input,

$$\mathcal{L}_{recon} = \|\hat{B} - B\|^2 \quad (1)$$

where $(\{m(B), E, A, M\})$ is input to the MLLM, and the latent MLLM representations associated with the BEV token embeddings are input to the masked reconstruction decoder head. Here, $m(\cdot)$ denotes random masking.

2) *Future BEV prediction:* An important aspect of planning is anticipating future events. We introduce the surrogate task of future BEV prediction to encourage the LLM to learn spatio-temporal cues useful for planning. Given latent BEV token embeddings, we train a prediction head to predict future BEV token embeddings and supervise this task as the L2 loss between predicted and ground truth future token embeddings:

$$\mathcal{L}_{future} = \|\hat{F}_t - B_{t+1}\|^2 + \|\tilde{F}_t - B_{t+2}\|^2 \quad (2)$$

where $\hat{F}_t, \tilde{F}_t$ are the predicted future BEV token embedding at time t . Note, we predict future BEV token embeddings (B_{t+1}, B_{t+2}) of the multi-view image sequence for time steps $\{t+1, t+2\}$.

3) *Scene editing:* For prediction and reasoning about the ego vehicle, it is crucial to learn how surrounding agents impact the ego-vehicle's future path. We propose a novel scene editing task in which we augment scenes by removing or adding new agents. Along with this, we construct a question-answer pair related to the edit. We show examples of this in Figure 3. For scene addition, given the map constraints, predicted map, the ego bounding box location and the trajectories of predicted agents, we create a way-point trajectory for a new agent of category "car" or "truck". A

new agent token embedding is then created using a linear layer. This new agent token embedding, the corresponding text prompt, and the rest of the *BEAM* token embeddings are passed as input to the LLM. The hidden latent LLM features corresponding to agent token embeddings are then fed into a dedicated scene editing decoder head that performs waypoint prediction of the ego vehicle. The language prediction head performs question-answering on the new QA pair. This task thus contributes to the existing planning constraint loss and VQA loss of the MLLM.

Distillation In order to further align the representations learned by the vision planner and the MLLM, we perform knowledge transfer between the penultimate layers of the language model and the planning transformer. Concretely, we minimize the KL-divergence between the distributions of the hidden features of the planning transformer and the LLM hidden features of the ego-token embeddings:

$$\mathcal{L}_{distill} = D_{KL}(P_{llm}||P_{vis}) \quad (3)$$

where the P_{vis} are the features of penultimate layers of vision planning transformer and P_{llm} are the hidden LLM embeddings features corresponding to the ego-token embeddings of penultimate layers of MLLM model.

Loss functions. We supervise the network with a weighted sum of the losses corresponding to planning, visual question answering, distillation, and each surrogate task.

$$\mathcal{L} = \mathcal{L}_{planning} + \mathcal{L}_{LLM} + \mathcal{L}_{recon} + \mathcal{L}_{future} + \mathcal{L}_{distill} \quad (4)$$

here $\mathcal{L}_{planning}$ comes from the planning objective used to train [18, 20]. Here, the loss weights are chosen to bring each value to the same scale.

4. Experimental Setup

In Section 4.1, we present the architecture design details, in Section 4.2 we present the training strategy, and in Section 4.3, we provide information about the datasets used. Importantly, in Section 4.4 we detail multiple evaluation strategies for fair and comprehensive comparison with existing works.

4.1. Architecture design

Vision-based planner We conduct experiments using two vision-based end-to-end planners, VAD [20] and UniAD [18] due to their performance, efficiency, and ability to model the interaction between scene components. We use two model size variants of VAD. Both planners are trained for perception, motion prediction and planning, while UniAD also performs occupancy prediction.

MLLM design The MLLM is made up of adapter layers, a language model, and a set of task-specific decoder layers. We project each *BEAM* token embedding with its own dedicated Q-former adapter layer following [17]. After projection, the *BEAM* token embeddings and language token embeddings lie in the same embedding space. In order to reduce memory consumption, we limit the agent token se-

quence length to a fraction of the total number of agents and perform sequence-wise upsampling after input to the language model. The adapter layers are randomly initialized. We use the LLM from LLaVA-v1.5-7B [24] as our language model base. The ego prediction and agent motion prediction task heads are multi-layer-perceptron networks following [20]. For the surrogate task decoder heads, we use 3 Linear layers with a ReLU activation layer.

4.2. Training

DiMA training follows a two-stage approach. First the vision-only planner is pre-trained for perception, prediction, and planning for 60 epochs in order to learn informative latent scene representations. Second, we perform joint training of the vision planner and the MLLM for an additional 30 epochs, incorporating all proposed tasks and losses detailed in Section 3. In the second stage, the language model of the MLLM is fine-tuned using LoRA [15]. Please refer to the supplementary material for more details.

4.3. Datasets

We use the nuScenes dataset for the task of open-loop planning [2]. This dataset consists of 28k total samples in a 22k/6k training/validation split. The objects in each scene are annotated with 3D bounding box, orientation, and vehicle speed information. Additionally, we make use of the CAN bus information for the ego vehicle’s ground-truth trajectory annotations. For supervising visual question answering, we train with DriveLM [37]. This dataset consists of a 4k subset of samples from the nuScenes dataset annotated with 300k QA pairs related to perception, prediction, planning, and behavior of the ego vehicle. For the samples in nuScenes that do not have text annotations, we create a set of perception, planning, and prediction QA pairs based on the numerical annotations. We use this information to prompt a Llama3-70B model [12] with in-context examples to generate DriveLM-like QA pairs. For the behavior QA pair, we create a rule-based algorithm to categorize the future movement of the ego-vehicle based on the ground-truth trajectory values and ego vehicle speed. More details on this method as well as examples of generated QA pairs may be found in the supplementary.

4.4. Evaluation details

We evaluate planning performance based on the predicted future trajectories of the ego vehicle 3 seconds into the future, where 2 waypoints are predicted per second. We use two metrics, the L2 error (in meters) between the predicted and ground-truth waypoints as well as the collision rate (in %) between the ego-vehicle and the surrounding vehicles.

Standardized vs VAD evaluation. In [48], Weng *et al.* point out inconsistencies in planning evaluation across VAD [20], UniAD [18] and AD-MLP [54], namely in the way L2 error is averaged over time and the way noisy and invalid

Table 1. Comparison of L2 trajectory error and collision rate on nuScenes [2] using standardized evaluation [48]. Models are evaluated on the general validation split as well as a “targeted” split of challenging samples from [48]. The performance of the DiMA model variants are in shades of purple. We summarize results by averaging over at $t = \{1, 2, 3\}s$ as well as at all time steps. The best average performance in each setting is in bold. “+” indicates the use of ego-status information.

Method	Using	Traj L2 (m) ↓					Collision (%) ↓				
	Ego status	1s	2s	3s	Ave _{1,2,3s}	Ave _{all}	1s	2s	3s	Ave _{1,2,3s}	Ave _{all}
Full validation split											
UniAD[18]	✗	0.48	0.89	1.47	0.95	0.83	0.32	0.29	0.73	0.45	0.40
VAD-Base [20]	✗	0.41	0.86	1.46	0.91	0.78	0.02	0.26	0.83	0.37	0.30
PARA-Drive[48]	✗	0.26	0.59	1.12	0.66	0.56	0.00	0.12	0.65	0.26	0.17
TOKEN[38]	✗	0.26	0.70	1.46	0.81	0.68	—	—	—	—	0.15
DiMA (VAD-Tiny)	✗	0.20	0.53	1.10	0.61	0.51	0.00	0.07	0.19	0.09	0.06
DiMA (VAD-Base)	✗	0.18	0.50	1.02	0.57	0.47	0.00	0.05	0.16	0.07	0.06
AD-MLP[54]	✓	0.23	0.58	1.18	0.66	0.56	0.00	0.14	0.70	0.28	0.20
PARA-Drive+[48]	✓	0.20	0.52	1.04	0.59	0.49	0.00	0.09	0.49	0.19	0.13
DiMA+ (VAD-Tiny)	✓	0.19	0.51	1.06	0.59	0.49	0.00	0.06	0.16	0.08	0.06
DiMA+ (VAD-Base)	✓	0.18	0.48	1.01	0.56	0.46	0.00	0.05	0.16	0.07	0.06
DiMA-Dual+ (VAD-Tiny)	✓	0.18	0.50	1.03	0.57	0.47	0.00	0.05	0.16	0.08	0.06
Targeted validation split											
UniAD[18]	✗	0.47	1.09	1.92	1.16	0.99	0.00	0.00	0.73	0.24	0.15
VAD-Base [20]	✗	0.53	1.20	2.07	1.27	1.08	0.00	0.29	0.87	0.39	0.34
PARA-Drive[48]	✗	0.38	0.97	1.88	1.08	0.91	0.00	0.00	0.72	0.24	0.14
DiMA (VAD-Tiny)	✗	0.32	0.88	1.70	0.97	0.81	0.00	0.03	0.24	0.09	0.05
DiMA (VAD-Base)	✗	0.29	0.79	1.49	0.83	0.71	0.00	0.03	0.22	0.08	0.05
AD-MLP[54]	✓	0.33	0.99	2.06	1.13	0.94	0.00	0.58	3.62	1.40	0.94
PARA-Drive+ [48]	✓	0.29	0.76	1.46	0.83	0.70	0.00	0.00	0.29	0.10	0.05
DiMA+ (VAD-Tiny)	✓	0.31	0.83	1.67	0.94	0.77	0.00	0.04	0.17	0.07	0.05
DiMA+ (VAD-Base)	✓	0.27	0.74	1.38	0.79	0.64	0.00	0.03	0.16	0.06	0.04
DiMA-Dual+ (VAD-Tiny)	✓	0.28	0.76	1.49	0.84	0.69	0.00	0.03	0.17	0.07	0.05

frames are considered. The authors also point out that collision performance is considerably improved by using a more finely discretized BEV map. Accounting for these inconsistencies, they propose a “standardized” evaluation metric to maintain a fair comparison across methods. For accurate reporting and fair comparison with [38, 48], we use this standardized metric to evaluate our model. Certain existing works use the evaluation scheme followed by [20] (which we call VAD evaluation) and do not make code or models available [16, 30, 39, 43]. Here, we compare reported results against DiMA evaluated using VAD evaluation.

Validation splits. We evaluate planning performance across 3 types of splits from the nuScenes validation set: 1) the entire validation set of 6019 samples 2) a 689 sample sub-set more difficult “targeted” scenarios in which the ego-vehicle turns during navigation [48], and 3) long-tail events in which the ego-vehicle encounters rare maneuvers [38]. u

5. Experimental Results

We perform a comprehensive evaluation of DiMA on the nuScenes open-loop planning benchmark in a variety of evaluation settings. We compare our performance against that of recent vision-based planners [16, 18, 20, 48] as well as recent MLLM planners [30, 38, 39, 43]. To facilitate a

fair comparison of DiMA with recent works, we use two evaluation strategies VAD evaluation [20] and standardized evaluation [48]. Please refer to Section 4.4 for more details.

Standardized evaluation. Under the standardized evaluation strategy, we evaluate each method on two validation splits: 1) the full nuScenes validation split, and 2) a “targeted” subset of the nuScenes validation split that contains scenarios where the ego-vehicle must take a turn, as established in [48]. We present our results in Table 1. The baseline vision planners VAD [20] and UniAD [18] exhibit sub-optimal performance in more difficult targeted scenarios. DiMA demonstrates significant improvements over baseline methods, achieving a 35% reduction in L2 trajectory error and an 80% reduction in collision rate on the overall nuScenes validation set, and a 35% reduction in L2 trajectory error in the “targeted” split. More notably, we outperform recent state-of-the-art methods PARA-Drive [48] and TOKEN [38] consistently across both validation splits. We observe a strong improvement in collision rate, improving on that of TOKEN by 60%. Unlike TOKEN, our approach does not rely on an LLM for inference, thus making DiMA more efficient as well as more accurate.

VAD evaluation. For a fair comparison with recent works that use the evaluation strategy from VAD [20], we present

Table 2. Comparison of L2 trajectory error and collision rate on nuScenes [2] using VAD evaluation [20]. Models are evaluated on the general validation split. The performance of the DiMA model variants are in shades of purple. The performance of the MLLM-branch DiMA model variant is in blue. We summarize results by averaging over all time steps. The best average performance in each setting is in bold. “+” indicates the use of ego-status information.

Method	Using Ego status	Traj L2 (m) ↓				Collision (%) ↓				Latency (ms)	FPS
		1s	2s	3s	Ave_{all}	1s	2s	3s	Ave_{all}		
ST-P3 [16]	✗	1.33	2.11	2.90	2.11	0.23	0.62	1.27	0.71	628.3	1.6
UniAD[18]	✗	0.48	0.96	1.65	1.03	0.05	0.17	0.71	0.31	555.6	1.8
VAD-Tiny[20]	✗	0.46	0.76	1.12	0.78	0.21	0.35	0.58	0.38	59.5	16.8
VAD-Base[20]	✗	0.41	0.70	1.05	0.72	0.07	0.17	0.41	0.22	224.3	4.5
VLP-VAD-Base[30]	✗	0.26	0.47	0.78	0.50	0.12	0.17	0.42	0.23	–	–
OmniDrive [43]	✗	0.40	0.80	1.32	0.84	0.04	0.46	2.32	0.94	–	–
DiMA (UniAD)	✗	0.15	0.30	0.56	0.34	0.06	0.08	0.22	0.12	560	1.8
DiMA (VAD-Tiny)	✗	0.18	0.36	0.61	0.38	0.07	0.10	0.27	0.15	59.5	16.8
DiMA (VAD-Base)	✗	0.13	0.27	0.47	0.29	0.05	0.08	0.16	0.10	226	4.5
VAD-Tiny	✓	0.20	0.38	0.65	0.41	0.10	0.12	0.27	0.16	59.5	16.8
VAD-Base	✓	0.17	0.34	0.60	0.37	0.07	0.10	0.24	0.14	224.3	4.5
DriveVLM [39]	✓	0.18	0.34	0.68	0.40	0.10	0.22	0.45	0.27	–	–
DriveVLM-Dual (VAD-Base) [39]	✓	0.15	0.29	0.48	0.31	0.05	0.08	0.17	0.10	–	–
DiMA + (VAD-Tiny)	✓	0.17	0.33	0.59	0.37	0.07	0.10	0.26	0.14	59.5	16.8
DiMA + (VAD-Base)	✓	0.12	0.25	0.44	0.27	0.04	0.06	0.15	0.08	226	4.5
DiMA-Dual+ (VAD-Tiny)	✓	0.14	0.27	0.46	0.29	0.05	0.07	0.15	0.09	286	3.5

Table 3. Long-tail performance comparison of L2 trajectory error and collision rate on nuScenes [2] validation set using standardized evaluation [48]. The performance of vision planner DiMA model variants are in shades of purple. The performance of MLLM-branch DiMA model variants are in shades of blue. We summarize results by averaging over at $t = \{1, 2, 3\}$ s as well as at all time steps. The best average performance in each setting is in bold.

Method	Traj L2 (m) ↓				Coll. (%) ↓	
	1s	2s	3s	Ave_{1,2,3s}	Ave_{all}	Ave_{all}
3-point turn (zero-shot)						
VAD-Tiny	0.76	1.68	3.04	1.83	1.55	0.00
VAD-Base	0.71	1.66	3.24	1.87	1.57	0.00
PARA-Drive [48]	0.50	1.38	2.76	1.55	1.29	5.33
TOKEN [38]	0.39	1.29	2.60	1.43	1.18	4.00
DiMA (VAD-Tiny)	0.47	1.27	2.55	1.43	1.17	0.00
DiMA (VAD-Base)	0.36	1.18	2.37	1.30	1.05	0.00
DiMA (MLLM)	0.38	1.12	2.35	1.28	1.04	0.00
DiMA-Dual (VAD-Tiny)	0.38	1.08	2.35	1.27	1.04	0.00
Resume from stop						
VAD-Tiny	0.64	1.63	2.99	1.75	1.49	0.00
VAD-Base	0.60	1.72	2.83	1.72	1.42	0.00
PARA-Drive	0.14	0.79	2.30	1.08	0.85	0.00
TOKEN	0.13	0.70	1.58	0.80	0.65	0.00
DiMA (VAD-Tiny)	0.26	1.02	2.40	1.23	0.99	0.00
DiMA (VAD-Base)	0.15	0.65	1.34	0.71	0.66	0.00
DiMA (MLLM)	0.24	1.02	2.16	1.14	0.93	0.00
DiMA-Dual (VAD-Tiny)	0.24	0.98	2.13	1.11	0.91	0.00
Overtake						
VAD-Tiny	0.58	1.27	2.12	1.32	1.14	2.42
VAD-Base	0.46	1.16	2.17	1.26	1.06	2.49
PARA-Drive	0.27	0.89	1.94	1.03	0.85	2.30
TOKEN	0.29	0.77	1.63	0.90	0.74	0.00
DiMA (VAD-Tiny)	0.24	0.75	1.49	0.83	0.69	1.32
DiMA (VAD-Base)	0.24	0.72	1.50	0.82	0.66	1.29
DiMA (MLLM)	0.24	0.73	1.50	0.82	0.67	1.30
DiMA-Dual (VAD-Tiny)	0.24	0.73	1.50	0.82	0.67	1.30

an additional set of results under this strategy.¹ In Table 2, we present the evaluation of DiMA on the nuScenes validation set [2]. We significantly outperform the base-

line vision-based planners VAD [20] and UniAD [18]. Notably, DiMA built on VAD-Tiny outperforms VAD-Base by 47% in L2 trajectory error while being 4 times faster. We also outperform MLLM planners like OmniDrive [43] and DriveVLM [39] without requiring an LLM at inference, making our approach more accurate as well as efficient.

Performance in long-tail scenarios. Autonomous driving performance is particularly crucial for rare scenarios. Using the manually selected long-tail events from [38], we demonstrate the robustness of DiMA in cases of novel navigation and perception. In Table 3, we evaluate the performance of DiMA for three long-tail scenarios, “3-point turn”, “resume from the stop” and “overtake”, selected from the nuScenes validation set. Training with DiMA results in a significant and consistent boost in performance from the baseline vision-based planners. We also consistently outperform PARA-Drive and TOKEN. Notably, the “3-point turn” case is a zero-shot scenario, not present in the training data. We achieve the lowest L2 trajectory error for this case.

Evaluating the MLLM branch. We also evaluate the MLLM branch of DiMA and demonstrate its planning and visual question-answering capabilities. We perform LLM-based planning using DiMA in two ways. First, we directly evaluate the waypoint predictions of the MLLM planning decoder head. This is denoted by DiMA (MLLM). Secondly, we define DiMA-Dual, an optional hybrid inference model that combines the strengths of both the vision and MLLM branches. We perform max-pooling on the penultimate layer features of both vision planning transformer and MLLM ego prediction. The resulting pooled features are then fed back into both the vision planning transformer and

¹We compare with reported results due to lack of published code.

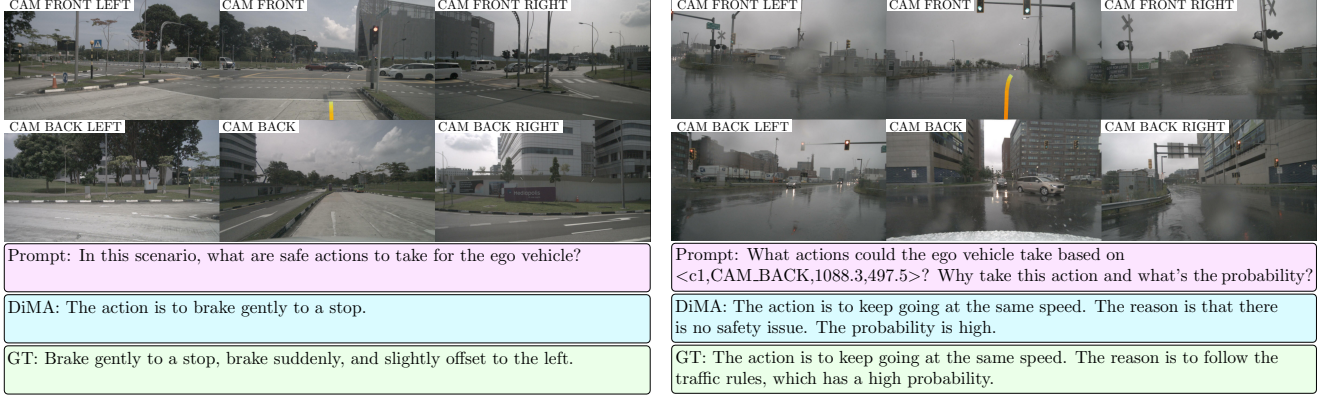


Figure 4. Visualization of planning performance by the MLLM branch of DiMA-VAD-Tiny. We plot the predicted trajectory (orange-yellow) and show an example response of the language model branch to a question from the DriveLM test dataset [37].

Table 4. Results of ablation experiments of DiMA-VAD-Tiny on nuScenes under VAD evaluation. The baseline model is VAD-Tiny. We report L2 trajectory error and collision rate averaged over all time steps.

ID	VQA	Scene tokens	Distillation	LLM Planning	Surrogate tasks			Traj L2 (m) ↓				Collision (%) ↓			
					Masked recon.	Future pred.	Scene editing	1s	2s	3s	Ave _{all}	1s	2s	3s	Ave _{all}
1	✗	✗	✗	✗	✗	✗	✗	0.33	0.58	0.89	0.60	0.12	0.19	0.55	0.29
2	✓	BEV	✗	✓	✗	✗	✗	0.36	0.60	0.91	0.62	0.10	0.17	0.52	0.26
3	✓	BEV, Map	✗	✓	✗	✗	✗	0.28	0.52	0.87	0.56	0.10	0.17	0.49	0.25
4	✓	All	✗	✓	✗	✗	✗	0.26	0.51	0.83	0.52	0.10	0.16	0.38	0.21
5	✓	All	✓	✓	✗	✗	✗	0.22	0.44	0.77	0.48	0.09	0.14	0.34	0.19
6	✓	All	✓	✓	✓	✗	✗	0.19	0.39	0.68	0.42	0.09	0.13	0.32	0.18
7	✓	All	✓	✓	✓	✓	✗	0.18	0.37	0.63	0.39	0.07	0.11	0.29	0.16
8	✓	All	✓	✓	✓	✓	✓	0.18	0.36	0.61	0.38	0.07	0.10	0.27	0.15

MLLM ego prediction for waypoint prediction. This max pooling encourages consistency between the predictions of the two models, leading to a more robust and potentially improved overall prediction as seen in Table 3. Using the MLLM at inference results in a significant boost in performance, with DiMA-Dual (VAD-Tiny) matching the performance of DiMA (VAD-Base) in Table 2.

VQA results. We provide qualitative examples of visual question-answering in the DriveLM dataset [37] in Figure 4. We demonstrate the ability of DiMA to reason about safe actions to be taken by the ego vehicle. We provide a more examples in the supplementary material.

5.1. Ablation study.

We examine the role of different design choices and proposed tasks in the DiMA framework through a comprehensive ablation study in Table 4. In order to do so, we build our final frame-work step-by-step. We begin with VAD-Tiny as our baseline model in the first row, denoted by ID-1. In ID-2, we observe inconsistent gains and drops in performance by naively training an LLM with the VAD-Tiny under a the planning and VQA objectives while using only BEV features. By comparing ID-3 and ID-4, we show that that providing more scene context to the MLLM by using all the *BEAM* token embeddings to train both LLM and

vision-based planner benefits the performance, resulting a significant improvement in trajectory estimation and collision rate. We show in ID-5 that explicitly distilling the information from the penultimate layers of the MLLM to the planning transformer gives another boost in performance. In ID-6, ID-7 and ID-8 we demonstrate the importance of surrogate tasks like masked-reconstruction, future prediction and scene editing in helping the scene encoder to learn richer representations useful for planning. We point to the supplementary material for additional ablation studies.

6. Conclusion

We introduce DiMA, an end-to-end autonomous driving framework for robust and efficient planning. DiMA distills knowledge from an MLLM to a vision-based planner, employing a novel joint training strategy alongside carefully designed surrogate tasks such as masked token reconstruction, future token prediction, and scene editing. DiMA enables the model to learn semantically grounded scene representations, improving its performance in difficult navigation scenarios. We conduct extensive evaluations and ablation studies, showing that training with DiMA results in a planner that excels in long-tail cases while maintaining computational efficiency.

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